

ACTIVE GRATING COOLING SYSTEM DESIGN FOR SOFT X-RAY MONOCHROMATOR

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Abstract

We present an active cooling system for the monochromator gratings of ID32 soft X-ray beamline at the European Synchrotron Radiation Facility (ESRF). The design combines flexible copper braids to minimize mechanical stress in the gratings with active temperature control to accelerate thermal response. Development followed a model-based approach, integrating dynamic Simulink thermal simulations with static finite element analysis. Under variable X-ray beam heat loads, the system maintains grating temperature stability within ± 2 mK and reduces thermal settling time from several hours to under 10 min. Interferometric measurements also confirm improved optical surface flatness, with the cooling system contributing less than 50 nrad RMS to slope error. This translates to improved beamline energy resolution from 25 meV to 22 meV at 930 eV photon energy.

INTRODUCTION

The ID32 Soft X-ray Beamline at ESRF consists of two branches: one dedicated to X-ray Absorption Spectroscopy (XAS) and the other to Resonant Inelastic X-ray Scattering (RIXS) studies [1]. Operational since October 2014, the beamline delivers polarized soft X-rays in the 400 eV to 1600 eV energy range. These are selected by a grating monochromator, which is the subject of this paper.

ID32 GRATING MONOCHROMATOR DESIGN

General Design

The ID32 grating monochromator was designed in collaboration between ESRF and BESTEC in 2011, based on the BESSY SX/700 concept [2]. The incoming white X-ray beam is first reflected by a cryogenically cooled pre-mirror, which absorbs most of the X-ray heat load. The deflected beam is then directed onto a water-cooled 200 mm long grating, producing a monochromatic X-ray beam in the 400 eV to 1600 eV range.

The monochromator is equipped with four interchangeable gratings: two for the XMCD branch of the beamline and two for the RIXS branch.

Each grating mirror is mounted on a separate pitch, roll and yaw adjustment support, while the cooling is ensured by a classical passive water-cooling circuit, featuring two side-cooling heat exchangers.

Limitations of Current Passive Grating Cooling System

ID32 users face performance limitations due to the grating assemblies, which rely on a conventional passive water-cooling circuit. The first limitation consists in the energy resolution, which is closely related to the flatness of the grating surface. Surface flatness is primarily influenced by the mechanical configuration of the cooling circuit.

The second limitation relates to the thermal settling time of the grating following variations in the input beam heat load. Energy drifts occur after significant changes to the Insertion Device (ID) settings, which can last up to several hours before the system reaches a steady state again.

These limitations were successfully addressed in a similar case by A. Crisol et al. through a novel active cooling approach featuring flexible heat-conducting foils and Peltier modules [3]. Inspired by this success, the present study explores a slightly different strategy.

ID32 ACTIVE GRATING COOLING SYSTEM DESIGN

Concept

The active grating cooling system designed by ESRF addresses the two above-mentioned limitations. Energy resolution is enhanced by improving the mechanical decoupling between the cooling system and the grating. This is achieved through the use of flexible heat-conducting copper braids between the heat exchangers and the water-cooling circuit, as well as a redesigned clamping mechanism for attaching the heat exchangers to the grating.

In a passive cooling configuration, the use of flexible heat-conducting copper braids would lead to a longer settling time due to the increased thermal resistance between the cooling water and the grating. However, this additional thermal resistance presents an opportunity to operate the heat exchangers in an active manner, as represented in Fig. 1.

In this approach, the heat exchangers not only cool the grating by transferring heat to the water circuit through the copper braids but also provide heating via an electric heater placed on the heat exchanger. This heater can supply heat to the grating, where the thermal resistance of the path remains relatively low, while heat loss to the water circuit is limited by the thermal resistance of the braids.

The electric heater operates within a feedback control loop, using a temperature sensor to regulate the grating temperature at a predefined setpoint. This setpoint is determined as follows: the system is subjected to the maximum input beam heat load, with a minimal yet nonzero contribution

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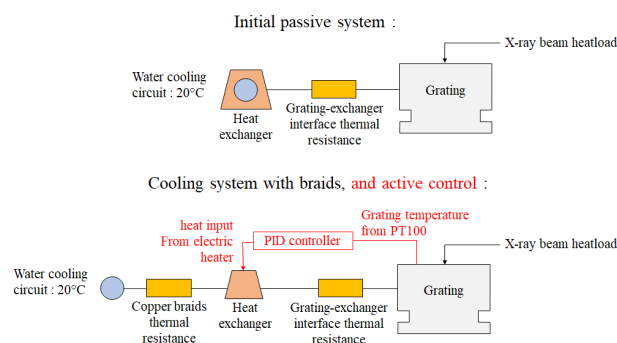


Figure 1: Electrical analogy of the effect of adding copper braids in the cooling path, and the opportunity of active control.

from the electric heater. The resulting steady-state temperature is then defined as the setpoint. In this configuration, if the X-ray beam heat load decreases, the control loop compensates by supplying additional power to the electric heater, maintaining the grating temperature at the desired setpoint.

Model-based Approach for Thermal Design

The design of the active cooling system for the gratings required consideration of several key aspects. One of the primary decisions involved selecting the appropriate temperature sensor technology and determining its optimal placement to ensure responsive temperature regulation. Another aspect was defining the location and power of the electric heater.

These decisions had to be made while accounting for the thermal resistance of the copper braids and the thermal mass of the grating. The objective was to achieve a settling time on the order of 10 min while maintaining a reasonable temperature setpoint below 305 K. Additionally, the design had to accommodate uncertainties, particularly regarding the thermal resistance between the heat absorber and the grating.

To guide these choices, a simplified dynamic model was implemented in MATLAB Simulink. This model enabled quick evaluations of different hardware configurations, temperature setpoints, and the impact of uncertainties related to the thermal resistance between the heat absorber and the grating.

Thermal Stresses in the Gratings

The hardware configuration chosen through the thermal design process raised several questions regarding heat distribution and the resulting mechanical stresses in the grating. The primary concern was the deformation of the grating due to localized heating from the electric heater. To assess this, an FEA model was developed in ANSYS under a defined worst-case X-ray beam heat load. The simulation demonstrated that heat from the electric heater is evenly distributed within the heat exchanger before being transferred to the grating, owing to the thermal resistance between the two

components. The resulting stresses and deformations in the grating have been stated as acceptable in this scenario.

Final Design

The pitch, roll, and yaw adjustment support from the original BESTEC grating holder design were retained, along with the original gratings, as represented in Fig. 2.

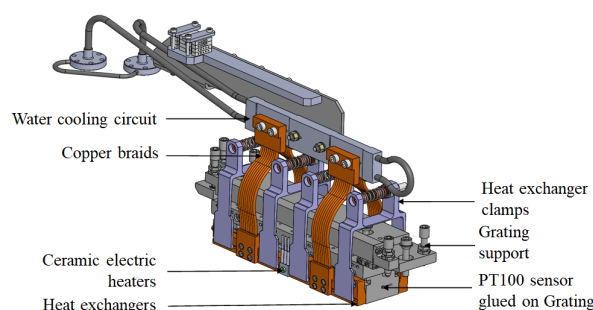


Figure 2: Overview of the active grating cooling system.

The heat exchangers were redesigned to integrate the electric heater, the copper braids, and the new heat exchangers clamping system. A 200 μm thick pure indium sheet was placed between the heat exchangers and the grating to enhance thermal contact.

Two copper heat exchangers are pressed against the grating using low-profile flexible clamps. Each clamp applies a controlled 30 N force, adjusted via a fine-threaded screw and a compression spring. To prevent mechanical coupling with other components, the clamps are solely supported by the heat exchangers.

Each heat exchanger is equipped with a ceramic electric heater, mounted with a single screw. The heaters are wired in series and connected to a single output port of the temperature controller. Although each heater has a maximum power rating of 100 W, the wiring limits the total power to 10 W per heater, which is considered sufficient based on the thermal design process.

Each heat exchanger is connected to the stainless-steel water circuit with two flexible heat-conducting copper braids manufactured by TAI. Each copper braid has a thermal resistance of 0.11 W K^{-1} . The water circuit is connected to the ESRF general cooling water supply, maintained at 20 $^{\circ}\text{C}$.

Temperature monitoring is ensured by two PT100 sensors bonded to the grating using UHV-compliant conductive epoxy glue (LOCTITE STYCAST 2850 FT). One sensor provides real-time feedback to the temperature controller, while the other is installed as a backup. The temperature controller used for this system is the LAKESHORE 340, a model already in use on the beamline and readily available for integration into the project.

Assembly and Grating Surface Measurements

One prototype of the active cooling system has been manufactured and tested with one of the four gratings in the monochromator.

The grating's reflecting surface was measured during assembly with a FIZEAU interferometer at the ESRF optical metrology laboratory. Compared with the previous passive system, improved decoupling of the water circuit and re-designed clamps greatly reduced forces transmission to the grating during assembly. The RMS residual profile error of the grating was measured at 200 nrad, with the cooling system contributing less than 50 nrad to the slope error.

BEAMLINE RESULTS

Commissioning tests of the system were conducted after installation of the active cooling system prototype on one of the four gratings in the monochromator.

Energy Stability Across Injections

One scenario affecting the input heat load is the gradual decrease in storage ring current between the hourly injections. Under these conditions, the PT100 sensor glued to the grating recorded a stable temperature within ± 2 mK. Low-frequency energy drifts in the range of 20 meV over several hours were observed and are likely attributable to the upstream pre-mirror, while the residual errors from the active cooling system contributing to higher-frequency energy oscillations did not exceed 1 meV as represented in Fig. 3.

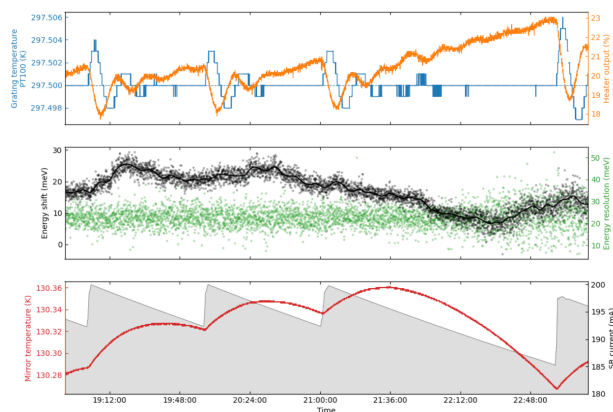


Figure 3: Photon energy stability across storage ring injections. Grating temperature and heater input (top), energy and energy resolution (middle), pre-mirror temperature and storage ring current (bottom).

Thermal Settling Time after Beam Loss

For a second test scenario, simulating a beam loss, the X-ray beam was shut off for 25 min before restoring it. In this case, settling time of the system was evaluated based on the time required to recover the initial energy resolution, which was approximately 8 min. A longer-term energy drift, extending over about 1 h, was also observed. This drift is most likely attributable again to the upstream pre-mirror, as illustrated in Fig. 4.

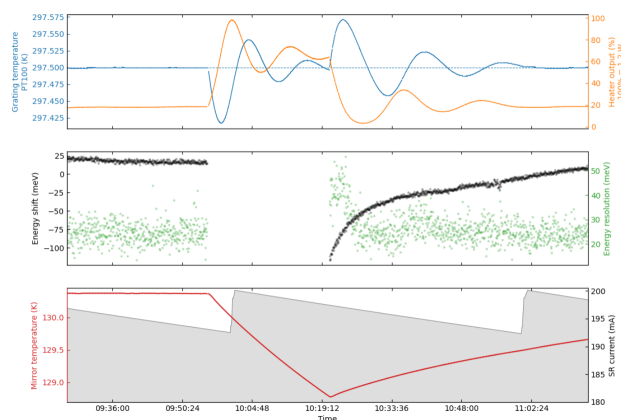


Figure 4: Photon energy stability after a beam loss. Grating temperature and heater input (top), energy and energy resolution (middle), pre-mirror temperature and storage ring current (bottom).

Impact on Energy Resolution

One of the main objectives in designing the active cooling system was to improve the beamline's energy resolution by minimizing mirror deformations during assembly. This improvement was also measured, considering the combined contributions of the grating mirrors, the spectrometer, and the pre-mirror. The overall energy resolution of the beamline improved from 25.6 meV to 22 meV by switching from the original passive cooling system to the newly implemented active cooling system on the grating mirrors. This is very close to the resolution expected for the free grating, considering only the intrinsic slope error of the grating surface.

CONCLUSION

An active grating cooling system developed by the ESRF has significantly improved both energy resolution and stability of the ID32 monochromator compared to the previously used passive cooling system. The energy resolution has improved from 25.6 meV to 22 meV, and the time during which energy drifts are observed after variations in the input beam heat load previously dictated by the thermal settling time of the grating, has been reduced from hours to just minutes.

While the current design already meets settling time requirements, future work focuses on further reducing the cooling system's contribution to grating slope error. A back-cooling concept combined with active temperature control is being explored with BESTEC as an innovative alternative to side-mounted exchangers.

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